

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Problem-Solving and Data Analysis

11

10 answers · Rationale-rich · Teach from doc

Q1**A****A. Increasing linear**

Choice A is correct. The graph of function f shows that as x increases, $f(x)$ also increases, which means $f(x)$ is an increasing function. The graph of f is a line, which indicates a constant rate of change. A function that has a constant rate of change is a linear function. Therefore, function f can be described as increasing linear.

Choice B is incorrect. For a decreasing function, as x increases, $f(x)$ decreases, rather than increases.

Choice C is incorrect. The graph of an exponential function isn't a line.

Choice D is incorrect. For a decreasing function, as x increases, $f(x)$ decreases, rather than increases, and the graph of an exponential function isn't a line.

Q2**D****D. 20**

Choice D is correct. It's given that the ratio of t to u is 1 to 2. Since $t = 10$, it follows that the ratio of 10 to u is also 1 to 2. The relationship between these ratios can be represented by the proportion $\frac{10}{u} = \frac{1}{2}$. Multiplying both sides of this equation by 2 and then by u yields $20 = u$.

Choice A is incorrect. This is the value of u when $t = 1$. Choice B is incorrect. This would be the value of u if the ratio of t to u were 2 to 1. Choice C is incorrect. This is the value of t , not u .

Q3**C****C.** 420

Choice C is correct. It's given that during an assembly, students occupied 50% of the 840 seats in the school auditorium. Therefore, the number of seats that the students occupied during this assembly can be calculated by multiplying the number of seats in the school auditorium by $\frac{50}{100}$. Thus, the students occupied $840 \frac{50}{100}$, or 420, seats during this assembly.

Choice A is incorrect. This is approximately 3%, not 50%, of 840.

Choice B is incorrect. This is approximately 6%, not 50%, of 840.

Choice D is incorrect. This is approximately 94%, not 50%, of 840.

Q4**B****B.** $\frac{2}{7}$

Choice B is correct. If a house from the street is selected at random, the probability of selecting a house that is blue is equal to the number of houses on the street that are blue divided by the total number of houses on the street. Since there are 2 blue houses on a street with 7 total houses, the probability of selecting a house that is blue from this street is $\frac{2}{7}$.

Choice A is incorrect. This is the probability of selecting a house that is blue from a street on which 1 of the 7 houses is blue.

Choice C is incorrect. This is the probability of selecting a house that is not blue from this street.

Choice D is incorrect. This is the probability of selecting a house that is blue from a street on which all the houses are blue.

Q5

A

A. The mean of the original data set is greater than the mean of the new data set.

Choice A is correct. It's given that the table shows an original data set of 5 values. It's also given that a sixth value is added to create a new data set. The new data set consists of the 5 values in the original data set and one additional value, 121. Since the additional value, 121, is less than any value in the original data set, the mean of the original data set is greater than the mean of the new data set.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

A

A. 2002

Choice A is correct. The year with the fewest graduates who enrolled in college within 24 months of graduation is the point with the lowest value on the vertical axis. This occurs at 2002.

Choice B, C, and D are incorrect. The years 2004, 2005, and 2007 each had a greater number of graduates who enrolled in college within 24 months of graduation than did the year 2002.

Q7**D**

D. 160

Choice D is correct. Since the ratio of y to x is constant for each ordered pair in the table, the first row can be used to determine that the ratio of y to x is 4 to 1. The proportion $\frac{4}{1} = \frac{k}{40}$ can be used to solve for k. Multiplying each side of the equation by 40 yields $160 = k$.

Choice A is incorrect. This is the value of y when the value of x is 7, not 40. Choice B is incorrect and may result from subtracting 4 from 40 instead of multiplying 40 by 4. Choice C is incorrect and may result from incorrectly setting up the proportion.

Q8**75**

The correct answer is 75. It's given that Amara harvested 750 cherry tomatoes and gave 10% of them to her neighbor. 10% can be written as $\frac{10}{100}$, or 0.1. Therefore, the number of cherry tomatoes that Amara gave to her neighbor can be calculated by multiplying the total number of cherry tomatoes by 0.1, which gives $(0.1)(750)$, or 75.

Q9**C**

c. $\frac{1}{2}$

Choice C is correct. If one of these planets is selected at random, the probability that the selected planet will be rocky is calculated by dividing the number of planets that are considered rocky by the total number of planets. It's given that 4 of the 8 total planets are considered rocky. Therefore, the probability that the selected planet will be rocky is $\frac{4}{8}$, which is equivalent to $\frac{1}{2}$.

Choices A and B are incorrect. These represent the probability if 1 of the 8 planets was considered rocky (choice A) and if 2 of the 8 planets were considered rocky (choice B). Choice D is incorrect and may result from dividing the total number of planets by the number of planets that are considered rocky.

Q10**8**

The correct answer is 8. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. It's given in the table that the times, in minutes, it took Robert to complete each of the tasks are 5, 2, 15, 9, and 9. The sum of these times is $5 + 2 + 15 + 9 + 9$, or 40, minutes. There are 5 tasks in the project. Therefore, the mean time, in minutes, for Robert to complete a task for this project is $40 \div 5$, or 8.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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